PROJECT REPORT

Retail Sales Analysis and AI-Based Sales Prediction

Submitted by

IBM SkillsBuild Data Analytics with AI Academic Internship Program

1. Abstract

This project presents a complete retail data analytics workflow combined with an AI-based sales prediction model. A reproducible synthetic dataset containing 1,500 retail transactions is generated for the year 2025. The project analyzes sales and profit patterns across product categories, regions, and customer segments. Visualizations are used to identify trends and business insights. A Random Forest Regression model is then trained to predict transaction-level sales using product category, region, customer segment, quantity, discount, and unit price. Model performance is evaluated using MAE, RMSE, and R^2 score.

2. Introduction

Data analytics helps organizations convert raw business data into useful information for decision-making. Retail organizations generate large volumes of transaction data, including product, customer, pricing, quantity, discount, sales, and profit information. This project demonstrates how Python-based analytics and machine learning can be used together to understand retail performance and predict sales.

3. Problem Statement

Retail businesses need to understand which categories, regions, and customer segments contribute to sales and profit. They also need predictive tools that can estimate expected sales for new transactions. The problem addressed in this project is to analyze retail transaction data and build an AI model that predicts sales from transaction attributes.

4. Objectives

• Perform data collection/generation and validation.

• Explore retail sales and profit patterns.

• Analyze category, region, and customer-segment performance.

• Visualize important trends and relationships.

• Build an AI/ML model for sales prediction.

• Evaluate the prediction model using standard regression metrics.

• Present the findings in a reproducible Jupyter/Python project.

5. Dataset Description

The project uses a reproducible synthetic dataset generated inside the code. It contains 1,500 retail transactions covering January 2025 to December 2025. The dataset includes Date, Category, Region, Customer_Segment, Quantity, Discount, Unit_Price, Sales, Profit, and Month. Synthetic generation was selected so that the complete project can be executed without relying on a restricted or unavailable external dataset.

6. Tools and Technologies

Python is used as the primary programming language. Pandas is used for data manipulation, NumPy for numerical operations, Matplotlib and Seaborn for visualization, and Scikit-learn for machine learning, preprocessing, train-test splitting, and evaluation. Jupyter Notebook can be used to execute and

document the workflow.

7. Methodology

The project follows these stages:

1. Import required libraries.

2. Generate a reproducible retail transaction dataset.

3. Check dataset shape, summary statistics, and missing values.

4. Perform exploratory data analysis.

5. Calculate business KPIs such as total sales, total profit, average order value, and profit margin.

6. Analyze sales by category and customer segment and profit by region.

7. Prepare categorical and numerical features for machine learning.

8. Split the dataset into training and testing sets.

9. Train a Random Forest Regression model using a preprocessing pipeline.

10. Evaluate the model using MAE, RMSE, and R^2.

11. Visualize actual versus predicted sales and feature importance.

12. Demonstrate a prediction for a new transaction.

8. Exploratory Data Analysis

The project includes monthly sales trend analysis, category-wise sales analysis, region-wise profit analysis, customer-segment summary statistics, and a correlation matrix. These analyses provide a structured view of business performance and show how different transaction variables relate to sales and profit.

9. AI Model

A Random Forest Regressor is used because it can model nonlinear relationships and interactions among transaction variables. Categorical variables are one-hot encoded using OneHotEncoder, while numerical variables are passed through directly. The complete preprocessing and model steps are combined in a Scikit-learn Pipeline to improve reproducibility.

10. Model Evaluation

The model is evaluated on a held-out test set using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. MAE represents the average absolute prediction error in sales units. RMSE gives greater weight to larger errors. R^2 indicates how much variation in the target is explained by the model. The exact values are produced automatically when the code is executed.

11. Business Insights

The project automatically calculates total sales, total profit, average order value, overall profit margin, top sales category, top profit region, and top customer segment. These outputs allow a business analyst to communicate measurable findings rather than relying on assumptions. Feature importance from the AI model also indicates which input variables contribute most to the prediction model.

12. Conclusion

This project demonstrates an end-to-end Data Analytics with AI workflow. It combines exploratory data analysis, visualization, business KPI calculation, and machine learning in one reproducible project. The Random Forest model provides a practical example of how AI can support sales estimation. The same workflow can be extended with real company data, dashboard development, advanced forecasting models, and additional customer/product features.

13. Future Scope

Future versions can use a real-world retail dataset, add customer lifetime value analysis, build monthly time-series forecasting, compare multiple machine-learning algorithms, and deploy the model through a Streamlit or Flask application. A Power BI dashboard can also be connected to the processed data for interactive reporting.

14. References

1. Python Documentation – https://docs.python.org/

2. Pandas Documentation – https://pandas.pydata.org/docs/

3. NumPy Documentation – https://numpy.org/doc/

4. Matplotlib Documentation – https://matplotlib.org/stable/

5. Seaborn Documentation – https://seaborn.pydata.org/

6. Scikit-learn Documentation – https://scikit-learn.org/stable/